Topic 1 Sets and Proofs

- 1.1 Sets of Numbers 2
- 1.2 Set Notations and Relations 8
- 1.3 Set Operations 35
- 1.4 Sets and Mathematical Proofs 57

1.1 Sets (of Numbers)

Unless you have taken some advanced studies, all of the calculus you have studied up until now has been about a particular class of *functions* - often called *functions of one variable* - that take a real number as input and give a real number as output.

In MAST10005, we will deepen our knowledge of the calculus of these functions and also start to extend our calculus studies a little beyond functions of one variable. We will also learn a little bit more about functions in non-numerical contexts.

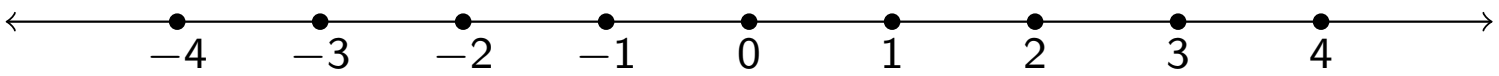
We begin by reviewing some sets we have seen before.

1.1 Sets of Numbers: Natural Numbers and Integers

- $\mathbb{N}$ is the set of natural numbers or counting numbers:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

- Whether or not to include 0 as a member of $\mathbb{N}$ is a matter of preference. (We will take the convention that 0 is not a member of $\mathbb{N}$.)
- $\mathbb{Z}$ is the set of integers or whole numbers[1]. It includes the positive and negative whole numbers as well as zero.



[1] The **Z** is for *Zahlen* – a German word for "number".

1.1 Sets of Numbers: Rational Numbers

- $\mathbb{Q}$ is the set of *rational numbers*[2] - those that can be written as fractions $\frac{m}{n}$ where m and n are integers, for example:

$$\frac{1}{2}, \frac{78}{23}, \frac{-3}{4}, \frac{0}{5} = 0, \frac{-6}{1} = -6, 1 = \frac{1}{1} = \frac{2}{2} = \frac{3}{3}, \ldots$$

- Every integer is a rational number.

[2] The **Q** is for *Quoziente* – Italian for "quotient".

1.1 Sets of Numbers: Irrational Numbers

- Some numbers cannot be written as fractions! We say they are irrational (i.e., not rational)
- A few examples:

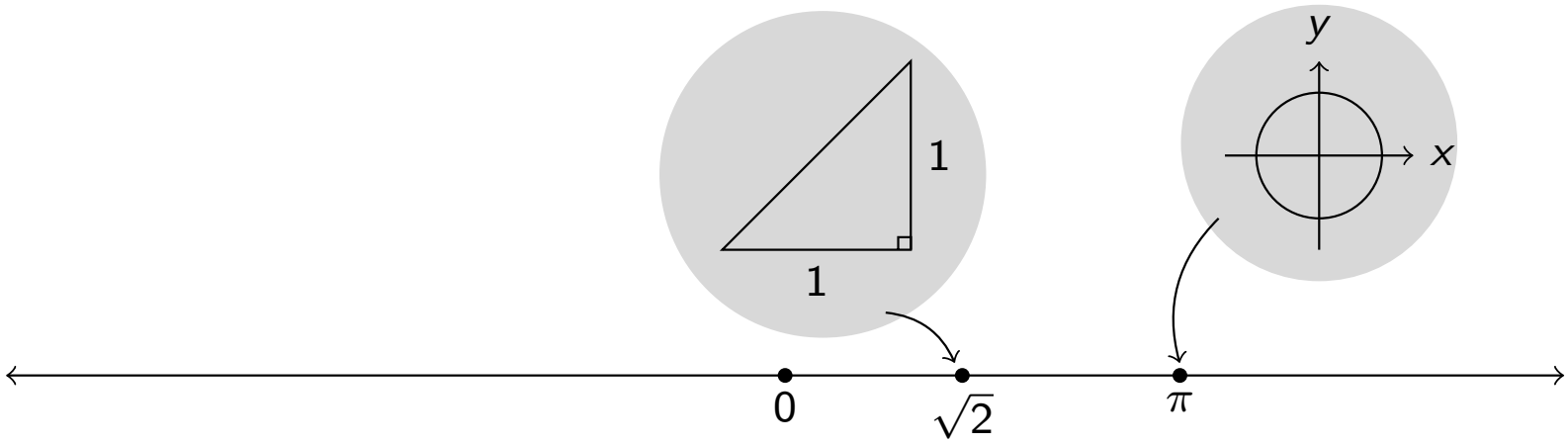
$$\pi, e, \sqrt{2}, \sqrt[3]{2}, \log_2(5), \log_5(2), \ldots$$

- In an upcoming tutorial we will *prove* irrational numbers exist.

... of course we will also have to deal with the the meaning of the word *prove* at some point as well...

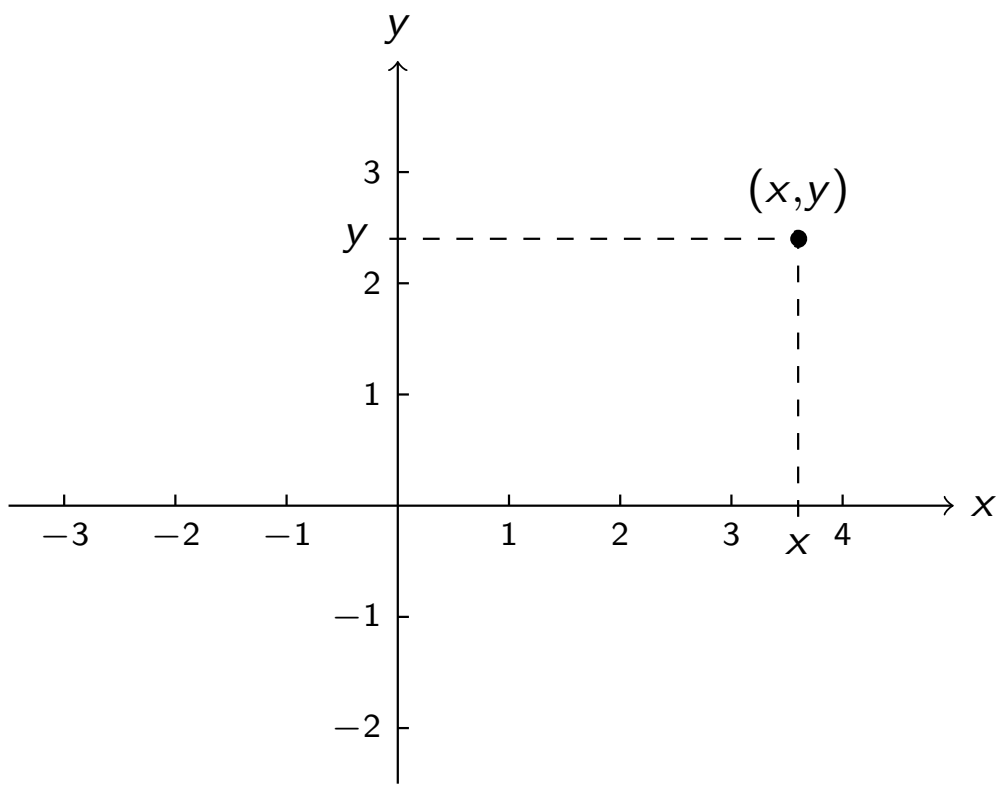
1.1 Sets of Numbers: The Real Numbers

- The set of irrational numbers joined together with the set of rational numbers gives the set of real numbers, which we write as $\mathbb{R}$.



1.1 Sets of Other Mathematical Objects

- Later we will work with many sets that are not subsets of $\mathbb{R}$. E.g.:
  - The complex numbers.
  - Higher dimensional sets like the plane and 3D space.



1.2 Set Notation and Relations

Throughout this subject we will define many different mathematical notations. Mathematical notation is used to ease written communication in mathematics. Usually, mathematical notation stands in for more complicated phrase.

1.2 Set Notation and Relations

e.g.,

the slope of the tangent line at a point to the function, where the output is given by the square of the input, is given by twice the x coordinate.

$$\frac{d}{dx}x^2 = 2x$$

1.2 Set Notation and Relations: What are sets?

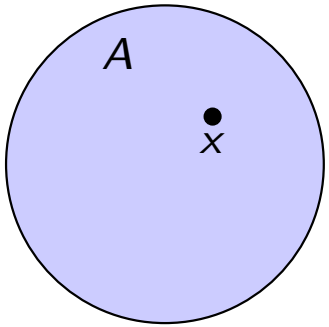
Before we go too far, let us take a moment to discuss what we actually mean when we use the word *set*.

A <u>set</u> is a collection of unique objects. A particular set is defined exactly by its members. For example, the set of students in this room.

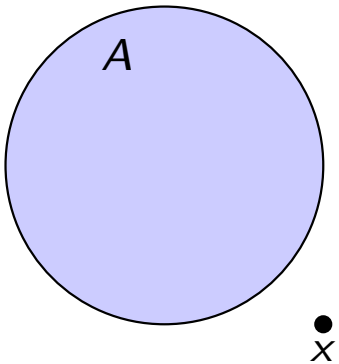
It turns out that giving a more precise definition of a set is hard. At some point during your mathematics education you might be tempted to consider the *set of all sets*. Thinking too hard about this concept will reveal why our definition above has some shortcomings. Ask your tutor if you are interested in hearing more!

1.2 Set Notation and Relations: Set Membership

- If A is a set, saying $x \in A$ means x is in the set A.



- When we write $x \in A$ we *say*:
  - "x is an element of A" or
  - "x is a member of A" or
  - "x is in A" or
  - "A contains x".
- We express the fact the statement $x \in A$ is false by writing $x \notin A$.



- When we write $x \notin A$ we *say*:
  - "x is not an element of A" or
  - "x is not a member of A" or
  - "x is not in A" or
  - "A does not contain x".
- Venn diagrams help us visualise set relationships.

1.2 Set Notation and Relations: Set Membership Examples

Membership examples:

- Writing $n \in \mathbb{Z}$ means n is an integer (positive, negative or zero).
- Writing $x \in \mathbb{Q}$ means x can be written as a fraction, so $x = \frac{m}{n}$ where $m \in \mathbb{Z}, n \in \mathbb{Z}$ with $n \neq 0$.
- Writing $y \in \mathbb{R}$ means y is some real number – it can be found somewhere on the real number line.

Non-membership examples:

- $\frac{3}{2} \notin \mathbb{Z}$ means $\frac{3}{2}$ is not an integer.
- $\log_2(5) \notin \mathbb{Q}$ means $\log_2(5)$ is not rational (i.e., it is irrational).
- $-1 \notin \mathbb{N}$ means -1 is not a natural number.

1.2 Set Notation and Relations: Defining a Particular Set

A set is defined precisely by its elements. We can do this by

- explicitly listing out the elements:

$$A = \{\text{dog}, \text{cat}, \text{koala}\}$$
$$B = \{0, 2, 4, 6, \dots\}$$

- or by giving a description of the set: *let B be the set of non-negative even numbers*,

1.2 Set Notation and Relations: Set Builder Notation

- or by using <u>set builder notation</u>.

For example, writing

$$A = \{x \in \mathbb{R} \mid x^2 + 1 > 37\}$$

defines the set A to be:

"the set of all real numbers x such that $x^2 + 1 > 37$ is true."

$$\begin{array}{ccccccc} A & = & \{ & x \in \mathbb{R} & \mid & x^2 + 1 > 37 & \} \\ & & \uparrow & \uparrow & \uparrow & \uparrow & \\ & & \text{The set of} & \text{all real numbers} & \text{such that} & \text{this statement is true.} & \end{array}$$

1.2 Set Notation and Relations: Set Builder Notation Examples

Example 1.1

1. Express *the set of real numbers whose natural (base e) logarithm is positive* in set builder notation.
2. Express *the set of integers whose cube is even* in set builder notation.
3. Describe the set $\{n \in \mathbb{N} \mid \sin(n) > 0\}$ with an English sentence.
4. Give some examples of elements of the set $A = \{x \in \mathbb{R} \mid \text{there exists } k \in \mathbb{Z} \text{ such that } x = k\pi\}$

Whether to use $\log_e(x)$, $\log(x)$ or $\ln(x)$ to refer to the natural logarithm is a matter of preference. In different mathematical communities, different choices are made. In the mathematical community of this class, we will opt for $\log(x)$.

1.2 Set Notation and Relations: Simplifying Set Builder Notation

- We can abbreviate set builder notation.
  For example, instead of writing
  $A = \{x \in \mathbb{R} \mid \text{there exists } k \in \mathbb{Z} \text{ such that } x = k\pi\}$ we can write

- We can read the abbreviated notation part by part:

$$\begin{array}{ccccccc} A & = & \{ & k\pi & \mid & k \in \mathbb{Z} \} \\ & & \uparrow & \uparrow & \uparrow & \uparrow \\ & & \text{The set of} & \text{real numbers} & \text{such that} & \text{this statement} \\ & & & \text{of this form} & & \text{is true.} \end{array}$$

- Note the left term (between $\{$ and $|$) is neither true nor false.
  It shows the form of the elements.

1.2 Set Notation and Relations: Simplifying Set Builder Notation

Example 1.2

1. Express the set of odd integers in abbreviated set builder notation.
2. Express the set $\{x \in \mathbb{R} \mid \cos(x) = 0\}$ in abbreviated set builder notation.
3. Express the set $\{x \in \mathbb{R} \mid \sin(x) = -1\}$ in abbreviated set builder notation.

1.2 Set Notation and Relations: Simplifying Set Builder Notation

Homework 1

1. *Express the set of integers whose square is no greater than their cube in set builder notation.*
2. *Describe the set* $\{x \in \mathbb{R} \mid x^2 > x^3\}$ *in words.*
3. *Express the set* $\{x \in \mathbb{R} \mid \cos(x) = 1\}$ *in abbreviated set notation.*
4. *Express the set* $\{x \in \mathbb{R} \mid \cos(x) = -1\}$ *in abbreviated set notation.*

1.2 Set Notation and Relations: Intervals of Real Numbers

Consider the following sets

$$
\begin{aligned}
A_1 &= \{x \in \mathbb{R} \mid 1 < x < 2\} \\
A_2 &= \{x \in \mathbb{R} \mid 1 \leq x < 2\} \\
A_3 &= \{x \in \mathbb{R} \mid 1 < x \leq 2\} \\
A_4 &= \{x \in \mathbb{R} \mid 1 \leq x \leq 2\}
\end{aligned}
$$

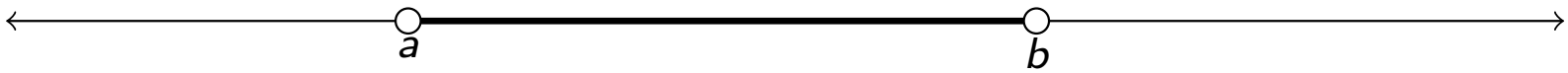
All of these sets are different. No two of them contain the exact same elements.

1.2 Set Notation and Relations: Intervals of Real Numbers

For $a, b \in \mathbb{R}$ with $a \leq b$:

- (a, b) means the interval of real numbers[3]

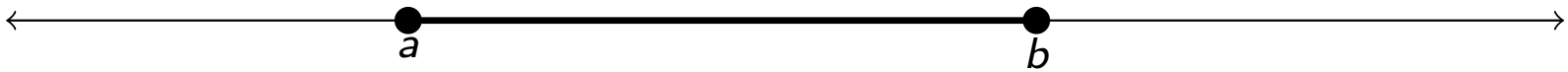
$$(a, b) = \{x \in \mathbb{R} \mid a < x \text{ and } x < b\} = \{x \in \mathbb{R} \mid a < x < b\}.$$



We refer to this interval as an *open* interval.

- $[a, b]$ means the interval of all real numbers from a to b, including a and b.

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \text{ and } x \leq b\} = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$



We refer to this interval as a *closed* interval.

[3] the same notation can mean different things in different contexts.

1.2 Set Notation and Relations: Intervals of Real Numbers

Example 1.3

Draw $[a, b)$ and $(a, b]$ on the real line.

1.2 Set Notation and Relations: Unbounded Intervals

Consider the set

$$A = \{x \in \mathbb{R} \mid x < 1\}$$

When we draw this set on set on the real line it *looks* like an interval. But it does not look like any of the intervals we have seen.

1.2 Set Notation and Relations: Unbounded Intervals

- For any $a \in \mathbb{R}$
  - $(-\infty, a)$ is the interval of real numbers strictly less than a.

$$(-\infty, a) = \{x \in \mathbb{R} \mid x < a\}.$$

  - (a, ∞) is the interval of real numbers strictly greater than a so

$$(a, \infty) = \{x \in \mathbb{R} \mid x > a\}.$$

- Note $-\infty$ and ∞ are not elements of $\mathbb{R}$. They are not numbers.

1.2 Set Notation and Relations: Unbounded Intervals

Homework 2

Give definitions of $(-\infty, a]$ *and* $[a, \infty)$*, with diagrams.*

1.2 Set Notation and Relations: The Empty Set

Consider the set

$$A = \{x \in \mathbb{R} \mid 1 < x < 1\}$$

1.2 Set Notation and Relations: The Empty Set

Every set with no elements is the *same*. We refer to the empty set using the notation $\emptyset$.

$$\{x \in \mathbb{R} \mid 1 < x < 1\} = \emptyset$$

1.2 Set Notation and Relations: Relating Sets

So far the notation we have defined for sets makes it more convenient to denote sets and their elements. But it gives us no tools to *relate* two sets.

We are well familiar with notation we can use to relate two numbers:

In each of these examples, the notation is standing in for statement about how two numbers relate to one another:

1.2 Set Notation and Relations: Set Equality

Are these two sets the same?

$$\{1, 2, 3, 4\} \qquad \{x \in \mathbb{N} \mid x < 5\}$$

How do you know?

1.2 Set Notation and Relations: Set Equality

Definition 1.4

Let A and B be sets. We say <u>A and B are equal</u> when every element of A is an element of B and every element of B is an element of A. When A and B are equal we write $A = B$.

We have <u>underlined</u> the thing we are defining. Now whenever we see $A = B$ from here on out, we should take it to mean *every element of A is an element of B and every element of B is an element of A.*

$$\{1, 2, 3, 4\} = \{x \in \mathbb{N} \mid x < 5\}$$

What does it mean for two real numbers to be equal?

1.2 Set Notation and Relations: Subsets

Every integer is a real number.

Every element of $\mathbb{Z}$ is an element of $\mathbb{R}$

$\mathbb{Z}$ is a subset of $\mathbb{R}$

If $x \in \mathbb{Z}$, then $x \in \mathbb{R}$.

$$\mathbb{Z} \subseteq \mathbb{R}$$

1.2 Set Notation and Relations: Subsets

Definition 1.5

Let A and B be sets. When every element of A is an element of B we say <u>A is a subset of B</u>. When A is a subset of B we write $A \subseteq B$.

When we write $A \subseteq B$, we *say*

- "A is a subset of B" or
- "A is contained in B" or
- "B contains A".

Whenever we see $A \subseteq B$ from here on out, we should take it to mean *every element of A is an element of B.*

Note: The empty set is a subset of every set. E.g., $\emptyset \subseteq \{1, 2, 3\}$

1.2 Set Notation and Relations: Subsets

$$\emptyset \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

Notice that the subset relation is *transitive*. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

1.3 Set Operations

So far we have developed notation that lets us denote whether or not an element is in a set, lets us describe a set and lets us compare two sets. However, like numbers, there is more we can do with sets.

We consider four operations:

1. set union
2. set intersection
3. set complement
4. Cartesian product of sets

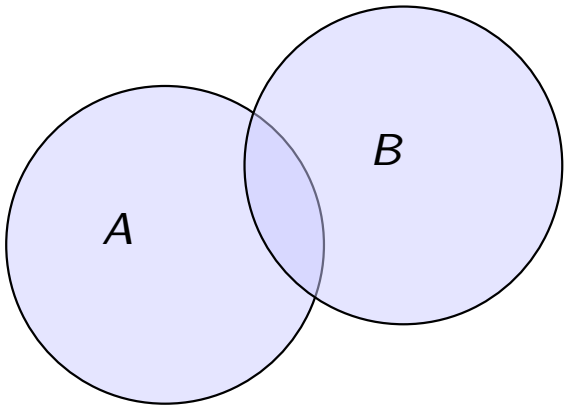
1.3 Set Operations

1.3 Set Operations: Set Union

Definition 1.6

Let A and B be sets. The <u>union of A and B</u> is the set of elements that are in at least one of A and B. We denote the union of A and B as $A \cup B$. That is

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$



Note that the use of *or* is inclusive in mathematics.

1.3 Set Operations: Set Union

Example 1.7

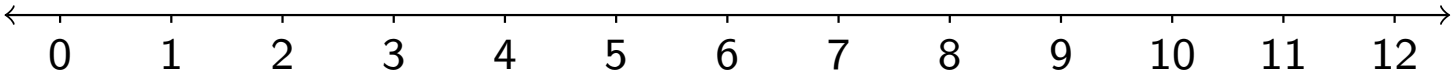
Express each of the following sets as a union of intervals.
Drawing graphs may help for the first two.

1. $\{x \in \mathbb{R} : x^2 > 1\}$.
2. $\{x \in (-2\pi, 2\pi] : \sin(x) \leq 0\}$.
3. $\{x \in [-2, 2] : x \notin \mathbb{Z}\}$.

Example 1.8

1. Express $(2, 8) \cup [3, 10]$ as an interval.
2. Is the set $(0, \sqrt{2}] \cup [\frac{\pi}{2}, 3)$ an interval?

1.



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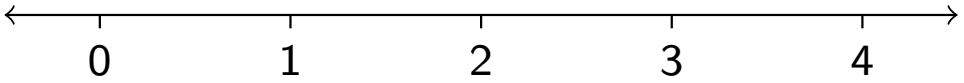


1.3 Set Operations: Set Union

Homework 3

Express the set $A = \{x \in \mathbb{R} \mid x^2 + 1 > 37\}$ *as a union of intervals.*

1.3 Set Operations

Let U be the set of undergraduate students at the University of Melbourne. Let C_1 be the set of all students enrolled in MAST10005 Calculus 1 this semester. Let C_2 be the set of all students enrolled in COMP10001 Foundations of Computing this semester.

There are some students who are in none of these sets, some students who are in at least one of these sets and some students who are in both of these sets.

- The set of students who are in at least one of MAST10005 and COMP10001 is denoted as $C_1 \cup C_2$.
- The set of students who are in both of MAST10005 and COMP10001 is denoted as $C_1 \cap C_2$.
- The set of students who are in neither of MAST10005 and COMP10001 is denoted as $U \setminus (C_1 \cup C_2)$.

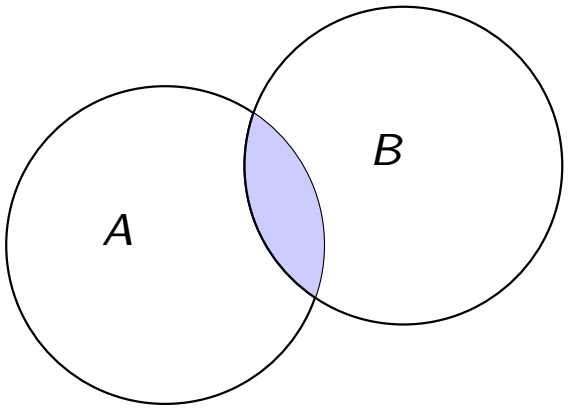
For this to make any sense to use we need to define the notation $\cup$, $\cap$ and $\setminus$.

1.3 Set Operations: Set Intersection

Definition 1.9

Let A and B be sets. The <u>intersection of A and B</u> is the set of elements that are in both of A and B. We denote the intersection of A and B as $A \cap B$. That is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

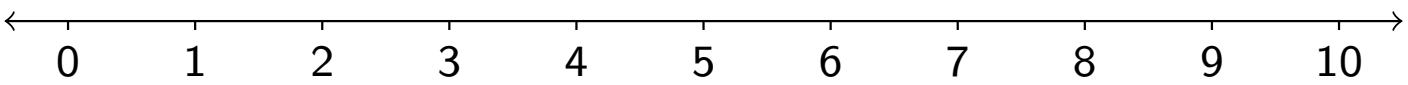


1.3 Set Operations: Set Intersection

Example 1.10

1. Express $(2, 8) \cap [3, 10]$ as an interval.
2. Express $(0, \sqrt{2}] \cap [\frac{\pi}{2}, 3)$ in the simplest possible way.
3. Express $\mathbb{Z} \cap [-\pi, \pi]$ as a list of elements.
4. Express $\mathbb{Z} \cap \{x \in \mathbb{R} \mid x^2 - 5 < 0\}$ as a list of elements.

1.



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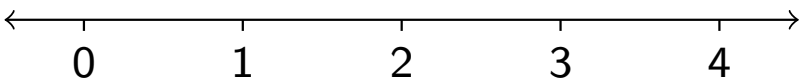
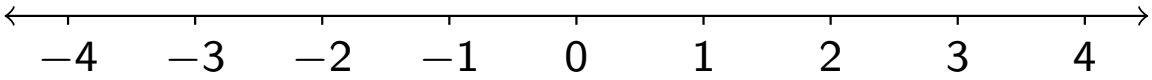


1.3 Set Operations: Set Intersection

3.



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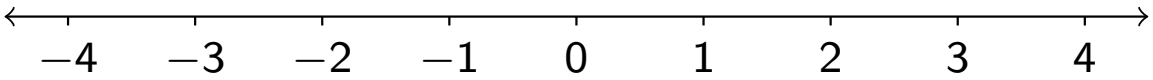


1.3 Set Operations

Example 1.11

Express the following sets as intersections and as single sets using set builder notation.

1. The set of reals with positive sine and negative cosine.
2. The set of integers with positive sine.

1.3 Set Operations

Let U be the set of undergraduate students at the University of Melbourne. Let C_1 be the set of all students enrolled in MAST10005 Calculus 1 this semester. Let C_2 be the set of all students enrolled in COMP10001 Foundations of Computing this semester.

There are some students who are in none of these sets, some students who are in at least one of these sets and some students who are in both of these sets.

- The set of students who are in at least one of MAST10005 and COMP10001 is denoted as $C_1 \cup C_2$.
- The set of students who are in both of MAST10005 and COMP10001 is denoted as $C_1 \cap C_2$.
- The set of students who are in neither of MAST10005 and COMP10001 is denoted as $U \setminus (C_1 \cup C_2)$.

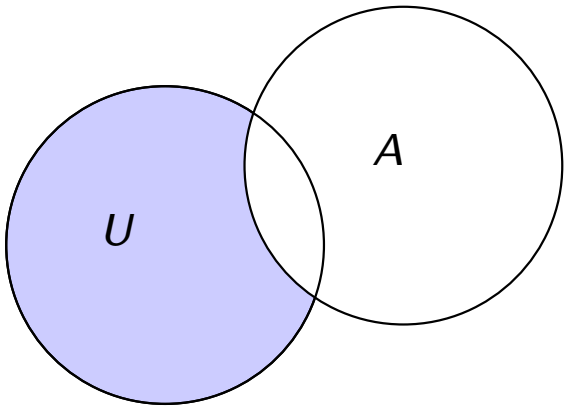
For this to make any sense to use we need to define the notation $\cup$, $\cap$ and $\setminus$.

1.3 Set Operations: Set Complement

Definition 1.12

Let U and A be sets. The <u>complement of A with respect to U</u> is the set of all elements of U that are not elements of A. We denote the complement of A with respect to U as $U \setminus A$. That is

$$U \setminus A = \{x \in U \mid x \notin A\}.$$

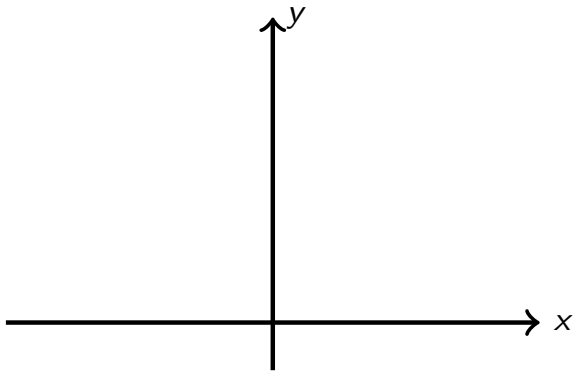


1.3 Set Operations: Set Complement

Example 1.13

Express each of the following sets as a set complement.

1. $\{x \in \mathbb{R} \mid x^2 > 1\}$.
2. $\{x \in [-2, 2] \mid x \notin \mathbb{Z}\}$.
3. $(-\infty, 0) \cup (0, \infty)$.



1.3 Set Operations: Set Complement

Homework 4

Express each of the following sets as a set complement.

1. $\{x \in [0, 2\pi] \mid \cos(x) > 0\}$.
2. $\{x \in \mathbb{R} \mid 1 - x^2 \leq x^2 - 1\}$.
3. *The set of irrational numbers.*

1.3 Set Operations: (Non)Commutativity

The real numbers have a very convenient property with respect to addition and multiplication. For every pair x and y of real numbers:

$$x + y = y + x \qquad \text{and} \qquad xy = yx$$

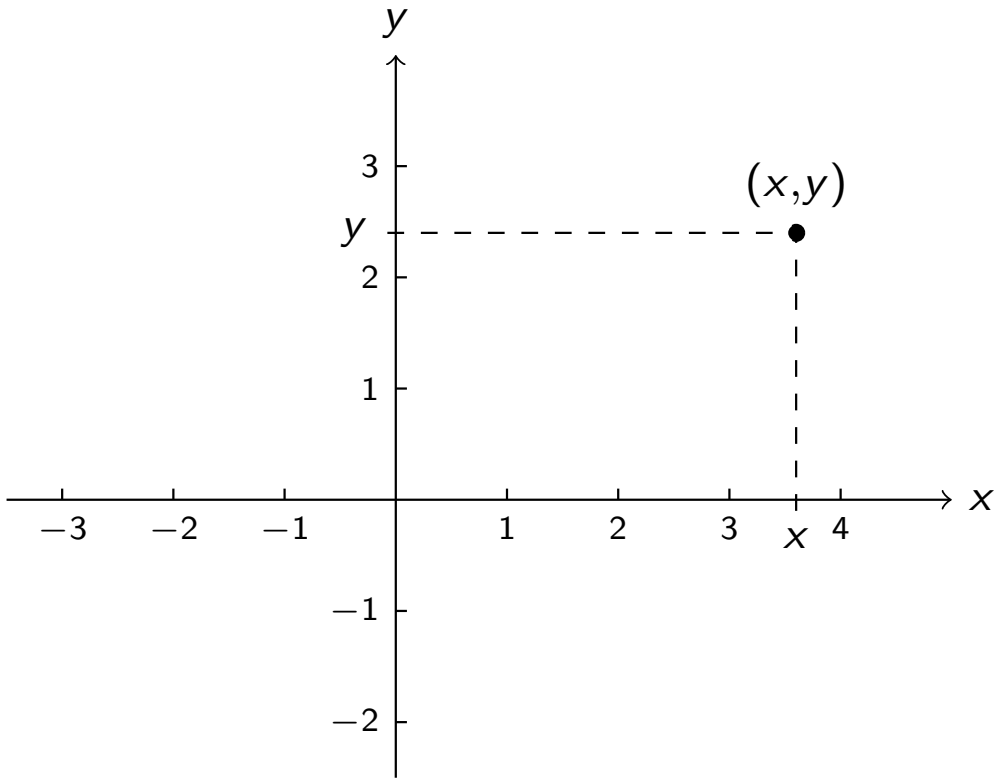
This is the <u>commutative</u> property.

Are each of the set operations commutative?

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$
$$U \backslash A = \{x \in U \mid x \notin A\}$$

1.3 Set Operations: Cartesian Product

Recall that we use $\mathbb{R}^2$ to refer to the points in the Cartesian plane.



$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

When $z \in \mathbb{R}$ we use z^2 to denote the product $z \times z$. The same thing is happening here: $\mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}$.

1.3 Set Operations: Cartesian Product

Definition 1.14

Let A and B be sets. The <u>Cartesian product of A and B</u> is the set of all possible *ordered* pairs we can build using elements of A as the first component and elements of B as the second component. We denote the Cartesian product of A and B as $A \times B$. That is

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

1.3 Set Operations: Cartesian Product

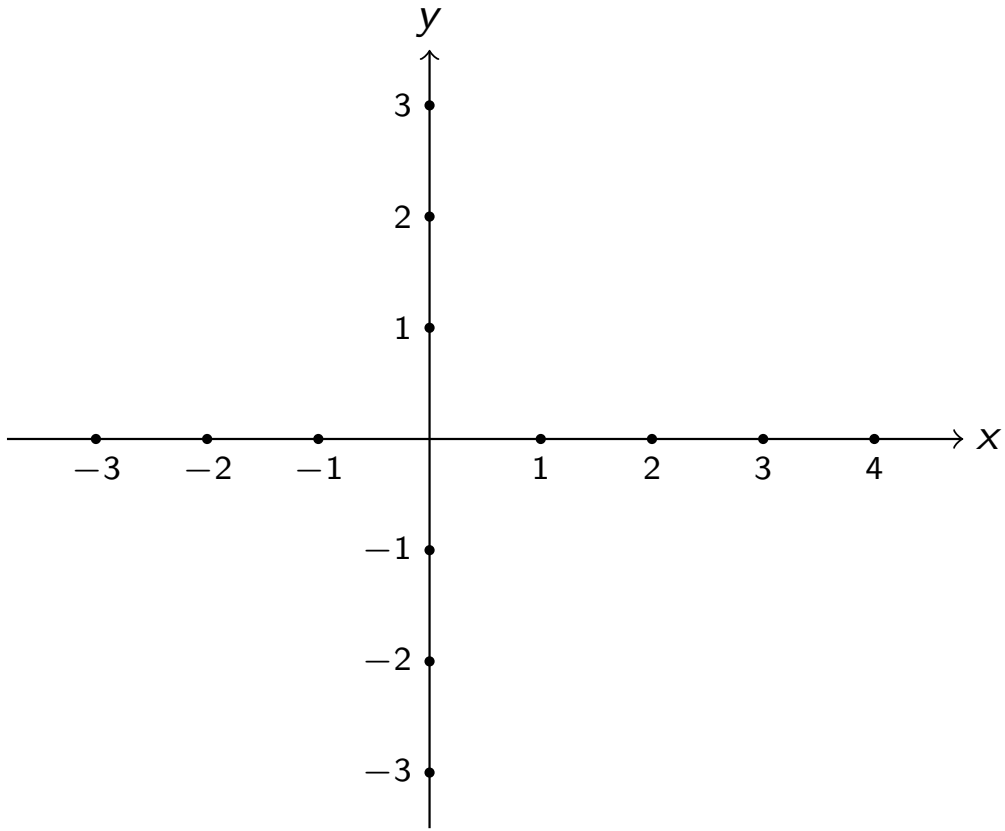
Example 1.15

List the elements of $A \times B$ where $A = \{0, 1, 5\}$ and $B = \{e, \pi\}$.

1.3 Set Operations: Cartesian Product

Example 1.16

Sketch the Cartesian product $\mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^2$ as a subset of $\mathbb{R}^2$.



1.3 Set Operations: Cartesian Product

Homework 5

Sketch the Cartesian products $\mathbb{Z} \times \mathbb{R}$ and $\mathbb{R} \times \mathbb{Z}$ as subsets of $\mathbb{R}^2$. Is the Cartesian product commutative?

1.4 Sets and Mathematical Proofs

Despite its reputation as a discipline of computation, one of the main activities for many mathematicians is proving results, as opposed to computing solutions.

1.4 Sets and Mathematical Proofs

Consider the following statement:

Let A, B and C be sets. If A is a subset of B and B is a subset of C, then A is a subset of C.

This statement is true, but how do we know?

1.4 Sets and Mathematical Proofs

Let us take a moment to look at the structure of this statement:

Hypothesis: A is a subset of B and B is a subset of C
Conclusion: A is a subset of C.

To verify the statement is true we want to assume that the hypothesis is true, and then use that information to deduce that the conclusion is true.

1.4 Sets and Mathematical Proofs

Let us look a little closer at the conclusion

Conclusion: A is a subset of C.

We defined the word <u>subset</u> as shorthand for a more detailed idea. Let us take a moment to find this back in our lecture notes.

To conclude that A is a subset of C, we need to show that every element of A is an element of C.

1.4 Sets and Mathematical Proofs

Goal: Consider $a \in A$. Conclude, somehow, that $a \in C$

Remember our hypothesis:

Hypothesis: A is a subset of B and B is a subset of C

- Let $a \in A$.
- Since A is a subset of B, then the definition of subset tells us $a \in B$.
- Since B is a subset of C, then the definition of subset tells us $a \in C$.

1.4 Sets and Mathematical Proofs

Theorem

Let A, B and C be sets. If A is a subset of B and B is a subset of C, then A is a subset of C.

Proof.

- Let $a \in A$.
- Since A is a subset of B, then by the definition of subset, $a \in B$.
- Since B is a subset of C, then by the definition of subset, $a \in C$.
- Therefore every element of A is an element of C.
- By the definition of subset, it then follows that A is a subset of C.

$\square$

1.4 Sets and Mathematical Proofs

Mathematicians use proofs to convince themselves and others of the truth of a mathematical statement. For the purposes of this subject, a good definition of proof is:

A <u>proof</u> is a sequence of true mathematical statements that convinces a peer with similar knowledge of the truth of a mathematical statement.

In our proof, we all agree on the definition of the notation we have used and the definition of subsets. And so if we accept the assumption: $A \subseteq B$ and $B \subseteq C$, then we must accept $A \subseteq C$

1.4 Sets and Mathematical Proofs

Theorem

If $C = \{(x, y) \in \mathbb{R}^2 \mid x^3 + y^2 = x^2\}$ *and* $P = \{(1 - t^2,\ t - t^3) \mid t \in \mathbb{R}\}$, *then* $P \subseteq C$

Though this is a very different looking statement to our previous example, it has the same form.

Hypothesis: $C = \{(x, y) \in \mathbb{R}^2 \mid x^3 + y^2 = x^2\}$ and
$P = \{(1 - t^2,\ t - t^3) \mid t \in \mathbb{R}\}$
Conclusion: $P \subseteq C$

1.4 Sets and Mathematical Proofs

Theorem

If $C = \{(x, y) \in \mathbb{R}^2 \mid x^3 + y^2 = x^2\}$ *and* $P = \{(1 - t^2,\ t - t^3) \mid t \in \mathbb{R}\}$, *then* $P \subseteq C$

Proof.

Let $C = \{(x, y) \in \mathbb{R}^2 \mid x^3 + y^2 = x^2\}$ and
$P = \{(1 - t^2,\ t - t^3) \mid t \in \mathbb{R}\}$.
Let $(x, y) \in P$.
This means $x = 1 - t^2$ and $y = t - t^3$ for some $t \in \mathbb{R}$. Hence:

$$\begin{aligned} x^3 + y^2 &= (1 - t^2)^3 + (t - t^3)^2 \\ &= 1 - 3t^2 + 3t^4 - t^6 + t^2 - 2t^4 + t^6 \\ &= 1 - 2t^2 + t^4 \\ &= (1 - t^2)^2 \\ &= x^2 \end{aligned}$$

Therefore $x^3 + y^2 = x^2$. Therefore $(x, y) \in C$. Therefore every element of P is an element of C. Therefore $P \subseteq C$ $\square$

1.4 Sets and Mathematical Proofs - Subset Proofs

From these two examples, we notice a pattern for proving one set is a subset of another.

Let A and B be sets
Hypothesis: Some information about A and B.
Conclusion: $A \subseteq B$.

Proof strategy: Consider an element $a \in A$. Use the information in the hypothesis as well as any relevant definitions and notation to deduce $a \in B$.

1.4 Sets and Mathematical Proofs - Subset Proofs

Example 1.17

Let A and B be sets such that $A = \{n \in \mathbb{N} \mid \sin(n) > 0\}$ and $B = \{n \in \mathbb{N} \mid \sin^2(n) \leq \sin(n)\}$. Prove $A \subseteq B$

1.4 Sets and Mathematical Proofs: Exploration

- When constructing a proof, we often reach a point where we do not know how to proceed. This is okay and very much expected. In such cases we must do some exploration.

Example 1.18

Let A and B be sets such that $A = \{4n \mid n \in \mathbb{Z}\}$ and $B = \{2m + 2 \mid m \in \mathbb{Z}\}$. Prove $A \subseteq B$.

Proof: Let $x \in A$. This means $x = 4n$ for some $n \in \mathbb{Z}$.

Exploration:

- Now we know the correct m value, we can continue with the formal proof argument:

1.4 Sets and Mathematical Proofs: Exploration

- **Important:** An exploration is not formally a part of the proof.
- It is like "rough working out". Anything goes!
- After we explore to find out what is needed, we must complete the proof by careful steps.

1.4 Sets and Mathematical Proofs: Counterexamples

So far all of our examples of proofs have been proving that a conclusion is true. However, not all mathematical statements are true!

$$\mathbb{Z} \subset \mathbb{N}$$

This statement is not true. How do you know?

1.4 Sets and Mathematical Proofs: Counterexamples

Let A and B be sets. To prove $A \subseteq B$ is a false statement, it suffices to find a single element of A that is not an element of B.

Example 1.19

Let $A = \{3n + 1 \mid n \in \mathbb{Z}\}$ and $B = \{6m + 1 \mid m \in \mathbb{Z}\}$. Prove $A \subseteq B$ is false.

1.4 Sets and Mathematical Proofs: Counterexamples

1.4 Sets and Mathematical Proofs: Contradictions

Let us take a closer look at the strategy we used to confirm $4 \notin B$. We assumed $4 \in B$, and then used that to conclude something absurd: there is a number that is simultaneously an integer and not an integer.

1.4 Sets and Mathematical Proofs: Contradictions

Prove $x^2 + x + 1 = 0$ has no real solutions.

Strategy: Assume that $x^2 + x + 1 = 0$ has a real solution, and then use that to deduce an absurdity.

1.4 Sets and Mathematical Proofs

Homework 6

Use the double angle formula for sine to prove
$\{x \in \mathbb{R} \mid \sin(x) = 0\} \subseteq \{x \in \mathbb{R} \mid \sin(2x) = 0\}$.

1.4 Sets and Mathematical Proofs

Homework 7

Prove that $\{x^3 \mid x \in \mathbb{R}\} \not\subseteq \{x^2 \mid x \in \mathbb{R}\}$.